DIGITAL PUBLIC INFRASTRUCTURE (DPI) — TECHNICAL REFERENCE SPECIFICATION

Federated Digital Assets Governance Model

A Privacy-by-Design Architecture for Semantic Interoperability, Automated Data Quality (ISO-as-Code), and Sovereign Information Mediation

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IP Registration	INDECOPI Res. N° 1990-2026/DDA	**Core Standards**	ISO 8000, ISO 25500, ISO 27001, W3C VC/DID
Target Ecosystem	DPI / Information Mediator Building Block	**Affiliation**	Independent Principal Data Architect (ISO 8000 / 25500 Certified)

1. Abstract & Architectural Problem Statement

Public sector digital ecosystems globally suffer from structural data fragmentation, semantic ambiguity, and excessive reliance on centralized data lakes. Consolidating sensitive citizen records into monolithic repositories creates single points of failure, violates data minimization principles, and introduces severe latency during multisectoral emergencies.

This specification defines the **Federated Digital Assets Governance Model**, an agnostic reference architecture designed to operate within the *Information Mediator* layer of Digital Public Infrastructure (DPI). Instead of replicating raw datasets, the architecture keeps data strictly at the source while enforcing automated, machine-executable data quality rules (**ISO-as-Code**) and cryptographic provenance verification across sovereign institutional nodes.

> **Capital Efficiency Principle (Zero-Licensing CAPEX):** The architecture eliminates capital expenditure on proprietary integration suites and redundant storage infrastructure. By leveraging open standards (W3C JSON-LD, OpenAPI, ISO 8000) and existing institutional databases, participating entities achieve cross-sector interoperability through semantic governance and infrastructure reuse optimization.

2. Regulatory & Global Compliance Alignment

The model is jurisdiction-agnostic and engineered to satisfy international data protection, cross-border flow, and public digital governance frameworks without compromising institutional autonomy:

- **Data Minimization & Privacy-by-Design:** Aligned with the European General Data Protection Regulation (GDPR Art. 5 & Art. 25) and the Council of Europe Convention 108+, ensuring zero raw Personally Identifiable Information (PII) is ingested or persisted at the central mediation layer.
- **Digital Public Goods & DPI Safeguards:** Adheres to the Digital Public Goods Alliance (DPGA) open-standard criteria and the UN Universal DPI Safeguards Framework regarding sovereignty, auditability, and algorithmic transparency.
- **Extraterritorial Data Sovereignty:** Enforces cryptographic policy gates at node boundaries, preventing unauthorized cross-border replication of sensitive population registries into non-equivalent jurisdictions.

3. Federated Governance & Ownership Architecture

In strict adherence to the DAMA-DMBOK framework, regulatory oversight is separated from operational data ownership. The central coordinating body never assumes ownership of institutional records; instead, accountability is distributed across three tiers:

Governance Tier	Designated Role	Core Mandate & Architectural Boundary
Tier 1: Ecosystem Orchestration	National Digital Governance Authority (NDGA)	Publishes the National Master Data Dictionary, maintains the canonical W3C JSON-LD ontology, and certifies Quality Gate rulesets. Zero access to raw transactional databases.
Tier 2: Source Node Custody	Source Data Owners (Institutional CDO / CAIO)	Retain legal custody and operational sovereignty over source systems. Responsible for curating compliant Data Products, signing provenance assertions, and enforcing local access policies.
Tier 3: Institutional Oversight	Digital Governance Committee	Cross-functional executive body (Operations, IT Architecture, Legal/DPO) authorizing node onboarding, risk thresholds, and budget allocation for local adapter maintenance.

4. Technical Standards Matrix (ISO-as-Code)

Data quality and supply chain integrity are enforced programmatically prior to payload exchange, translating international standards into deterministic validation rules:

Standard	Architectural Function	Executable Specification in the Mediation Layer
ISO 8000-110 Semantic Encoding	Canonical URI Resolution	Replaces ambiguous natural-language attributes with globally unique Uniform Resource Identifiers (URIs) bound to a machine-readable master ontology (ISO 8000-110).

Standard	Architectural Function	Executable Specification in the Mediation Layer
ISO 8000-120 Data Provenance	Cryptographic Lineage	Mandates explicit metadata headers recording originating institution DID, extraction timestamp, and transformation lineage for every federated query response.
ISO 8000-130 / 140 Accuracy & Completeness	Automated Quality Assertions	Generates verifiable digital assertions certifying syntactic validity, referential integrity, and completeness thresholds prior to cross-node consumption.
ISO 25500 Supply Chain Data	End-to-End Logistics Integrity	Standardizes master identifiers across public procurement, humanitarian logistics, and critical infrastructure distribution networks (ISO 25500).
ISO/IEC 27001 Information Security	Zero-Trust Node Transport	Enforces mutual TLS (mTLS), hardware-backed key signing, and immutable audit logging across all information mediator gateways.

5. Core Architectural Modules & Trust Layer

5.1. Module I: Canonical Ontology & Master Data Dictionary

Establishes a unified semantic layer where critical public entities (Citizen Identity Token, Legal Entity ID, Geospatial Parcel Code, Vulnerability Index) share a single machine-interpretable schema across sectors, eliminating costly point-to-point ETL translation pipelines.

5.2. Module II: Automated Quality Gate (ISO-as-Code Engine)

Replaces retrospective manual audits with a synchronous, edge-deployed validation gateway. Before an institutional node exposes a Data Product to the federated bus, the Quality Gate executes compiled rulesets verifying ISO 8000 syntactic/semantic conformance and PII masking. Non-compliant payloads are rejected at the source boundary with deterministic error codes.

5.3. Module III: Cryptographic Trust Layer (W3C VC / DID & Merkle Ledgers)

Rather than relying on energy-intensive public blockchains, the trust layer implements Decentralized Identifiers (W3C DIDs), Verifiable Credentials (W3C VC), and append-only Merkle Tree cryptographic hash logs (W3C VC Data Model). Only SHA-256 payload hashes, provenance timestamps, and quality assertions are committed to the distributed audit ledger, ensuring tamper-evident traceability with zero PII exposure.

5.4. Module IV: Sovereign Boundary & Synthetic Simulation Engine

Enforces automated geofencing and policy evaluation over data transit while providing a deterministic synthetic data generator. Policy makers and AI models can run stress tests and crisis simulations on statistically equivalent synthetic datasets without accessing real citizen records.

6. Reference Interface Contract (JSON-LD & ISO 8000 Payload)

To support direct integration within the GovStack *Information Mediator* Building Block, participating nodes exchange self-describing metadata envelopes structured in W3C JSON-LD. The following normative contract illustrates a zero-PII eligibility verification response:


```
{
  "@context": [
    "https://www.w3.org/2018/credentials/v1",
    "https://dpi-governance.org/ns/iso8000-mediator/v1"
  ],
  "id": "urn:uuid:8f4e2a10-9c3b-4d7e-b1a2-6e5f4c3d2b1a",
  "type": ["VerifiableCredential", "FederatedDataProductAssertion"],
  "issuer": "did:web:node.social-protection.gov.example",
  "issuanceDate": "2026-11-12T17:30:00Z",
  "iso8000Provenance": {
    "standard": "ISO-8000-120:2022",
    "sourceAuthority": "did:web:node.social-protection.gov.example",
    "semanticDictionaryURI": "https://dpi-
governance.org/ontology/v1#EmergencyReliefEligibility",
    "extractionTimestamp": "2026-11-12T17:29:58Z"
  },
  "iso8000QualityGate": {
    "standard": "ISO-8000-130-140:2022",
    "verifyStatus": "PASSED",
    "completenessScore": 1.00,
    "accuracyAssertionHash":
"sha256:9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08",
    "piiExposurePolicy": "ZERO_RAW_PII_ZK_ATTESTATION"
  },
  "credentialSubject": {
    "subjectTokenHash":
"sha256:4b227777d4dd1fc61c6f884f48641d02b4d121d3fd328cb08b5531fcacdabf8a",
    "eligibilityStatus": true,
    "vulnerabilityTier": "TIER_1_CRITICAL",
    "geospatialZoneCode": "ISO-3166-2:PE-LMA"
  }
}
```


7. Global Benchmarking & Architectural Differentiation

Reference Ecosystem	Core Strength	Federated Model Advancement
Estonia (X-Road)	Decentralized, cryptographically signed point-to-point data exchange.	Adds an automated **ISO-as-Code Quality Gate** that validates semantic accuracy (ISO 8000) before transmission.
Singapore (TradeTrust)	Verifiable electronic documents and provenance across trade networks.	Extends verifiable credentials to real-time public master data products and multi-agency emergency registries.

Reference Ecosystem	Core Strength	Federated Model Advancement
GovStack Mediator	Standardized API gateway and pub/sub routing for government building blocks.	Provides the normative **Data Governance & Quality Layer** missing from pure transport-level mediation.

8. Appendix: Reference Implementation Case Study (Climate Resilience)

To validate operational feasibility under high-stress conditions, the model was architected against a national climate emergency scenario (El Niño Phenomenon multisectoral response), coordinating four functional node domains without centralizing citizen databases:

- **Financial Continuity Node (Public Banking):** Executes real-time emergency subsidy disbursements using zero-PII cryptographic eligibility assertions, preventing duplicate payments and leakage.
- **Early Warning & Disaster Risk Node (Meteorological & Civil Defense):** Fuses dynamic hazard telemetry with sovereign municipal vulnerability registries using canonical geospatial URIs.
- **Humanitarian Supply Chain Node (ISO 25500):** Tracks critical medical and relief inventory end-to-end across regional distribution hubs with immutable Merkle audit proofs.
- **Public Health & Critical Infrastructure Node:** Monitors water, energy, and epidemiological indicators across affected jurisdictions while maintaining strict national custody over patient records.

9. Normative References

1. International Organization for Standardization. *ISO 8000 Series: Data Quality — Parts 110, 120, 130, and 140 (Master Data: Exchange of Characteristic Data, Provenance, Accuracy, and Completeness)*.
2. International Organization for Standardization. *ISO 25500 Series: Supply Chain Interoperability and Integration — Master Data Exchange Standards*.
3. DAMA International. *DAMA-DMBOK: Data Management Body of Knowledge (2nd Edition) — Federated Data Governance & Stewardship Frameworks*.
4. World Wide Web Consortium (W3C). *Verifiable Credentials Data Model v1.1 & Decentralized Identifiers (DIDs) v1.0 — Core Architecture and Data Formats*.
5. European Parliament and Council. *Regulation (EU) 2016/679 (General Data Protection Regulation) — Article 5 (Data Minimization) & Article 25 (Data Protection by Design and by Default)*.